Advanced Calculus I Hand-in Assignment 1

Student ID Number (學號): Name :

- This Assignment **will be collected in class.** (這份作業會在上課的時候收。) Please put your assignment on the desk in our classroom **before 10:25 on Sep 23th, Wednesday**. (請在9月23日10:25分以前，將作業放在教室的講台。) Please write down your student ID number, name and answer on A4 papers and staple your papers together （請將學號、姓名、答案寫在A4大小的紙張。若有兩張以上的答案紙，請用訂書機訂好。).
- Note that we do not collect late assignment. (請注意，不收遲交作業。)
- 作業在第2頁附上中文，希望可以在學習英文上對同學有所幫助，但是小考、期中、期末考不會附上中文，**建議同學先讀第一頁全英文的部份，再比較理解的情況和第二頁有什麼不同**。寫作業、考試可以寫中文或是中文、英文混合。

Problem 1. (10 points.) Suppose that $a, b,$ and $c \in \mathbb{R}$. Also, we have $0 < a < b$. Please show that

$$0 < \frac{1}{b} < \frac{1}{a}.$$

Problem 2. (10 points) Please show that there is no $n \in \mathbb{Z}$ that satisfies $0 < n < 1$.

Bonus Problem. (3 points) Let $E \subseteq \mathbb{R}$. We introduce the supremum of the set E in class and say that it is the least upper bound of E. In fact, we can define greatest lower bound of E by similar idea, which is called inimum of the set E. You may check how it is defined by our textbook. Please define the following terminology.
(1) E is bounded below. (1 point)
(2) A lower bound of the set E. (1 point)
(3) An infimum of the set E. (1 point)

Problem 1. (10 分.) 假設$a, b,$ 與$c \in \mathbb{R}$. 並且已知$0 < a < b$。請證明

$$0 < \frac{1}{b} < \frac{1}{a}.$$

Problem 2. (10 points) 請證明沒有任何整數n滿足$0 < n < 1$。

Bonus Problem. (3 points) Let $E \subseteq \mathbb{R}$. 我們在課堂上介紹了集合E的supremum，並在解釋了其意思就是集合E的最小上界。事實上，我們可以用類似的方式定義集合E的最大下界，其名稱爲集合E的infimum。同學可以參考教科書的內容，寫出下列名詞的定義
(1) 集合E有下界。(1 point)
(2) E的下界。(1 point)
(3) 集合E的infimum。(1 point)